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In-house project report
Mid Semester Examination March 2026
titled
**Virtual Screening and Drug design against Mycobacterium
Tuberculosis** completed
at
MITADTU School of Bioengineering Sciences & Research,
Ms. Shrutika Mane
Under the guidance of
Dr. Gourab Bhattacharjee
IN PARTIAL FULFILMENT FOR THE DEGREE
OF
M. Sc. Bioinformatics
Submitted to
School of Bioengineering Sciences & Research,
A Constituent Unit of MIT-ADT University, Pune.

2025 - 2026

CERTIFICATE

This is to certify that **Ms. Shrutika Mane (ADT25SBSM0167)** has been successfully completed in-house project entitled **“Virtual Screening and Drug design against Mycobacterium Tuberculosis”**, [GB_2025_PG_BI_002_MTB] under the guidance of **“Dr. Gourab Bhattacharje”** at School of Bioengineering Sciences & Research, a constituent unit of MIT-ADT University, Pune during the academic year 2025 -26, for the partial fulfillment for the award of the degree of M. Sc. Bioinformatics

Date: 29/04/2026

Place: PUNE

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DECLARATION

I declare that this report titled “Virtual Screening and Drug design against Mycobacterium Tuberculosis” [GB_2025_PG_BI_002_MTB] reflects my original work about the subject in my own words. I have sufficiently cited and referenced the original sources, referred or considered in this work. I have not plagiarized or submitted the same work for the award of any other degree. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute.

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ABSTRACT

Tuberculosis (TB), caused by *Mycobacterium tuberculosis*, remains a major global health challenge, further complicated by the emergence of multidrug-resistant (MDR) and extensively drug-resistant (XDR) strains. Conventional drug discovery approaches are time-consuming, costly, and often insufficient to keep pace with evolving resistance mechanisms. This project focuses on the application of computer-aided drug design (CADD) strategies, particularly virtual screening and structure-based drug design, to identify and optimize novel therapeutic candidates against *M. tuberculosis*.

The study involves the identification of key molecular targets essential for bacterial survival, followed by the retrieval and preparation of their three-dimensional structures. Large compound libraries are subjected to virtual screening to identify potential inhibitors with high binding affinity. Molecular docking techniques are employed to evaluate protein–ligand interactions and predict binding modes, while molecular dynamics simulations are used to assess the stability and behaviour of the selected complexes under physiological conditions. Furthermore, pharmacokinetic and toxicity profiles of promising compounds are analysed through ADMET prediction to ensure druglikeness and safety.

The project also explores the potential of drug repurposing and novel scaffold discovery to accelerate the development of effective anti-tubercular agents. By integrating computational tools, this study aims to identify promising lead compounds capable of overcoming drug resistance and improving treatment outcomes. Overall, the work demonstrates the efficiency and reliability of in silico approaches in streamlining the drug discovery process and provides a foundation for future experimental validation and development of new anti-TB therapies.

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Introduction and Literature review

1.1 Overview of Tuberculosis and Its Global Impact

Tuberculosis (TB), caused by *Mycobacterium tuberculosis*, continues to be one of the leading infectious causes of mortality worldwide. Despite the availability of effective chemotherapy, TB remains a major global health burden, particularly in developing countries. According to recent global health estimates, millions of new TB cases are reported annually, with a significant proportion progressing to active disease. The pathogen primarily affects the lungs but can also infect other organs (extrapulmonary TB). Its ability to persist in a latent state within host

macrophages makes eradication challenging. Additionally, socio-economic factors, co-infection with HIV, and inadequate healthcare infrastructure contribute to the continued prevalence of TB.

1.2 Historical Development of Anti-Tuberculosis Drugs

The treatment of tuberculosis historically relied on the discovery of several effective antimicrobial drugs during the mid-20th century. The introduction of antibiotics such as streptomycin (1944), para-aminosalicylic acid (PAS), isoniazid (1952), and pyrazinamide (1954) revolutionized TB therapy and significantly reduced mortality. Later, rifampicin and ethambutol were incorporated into standard multidrug therapy regimens.

These drugs function by targeting essential biological processes of *M. tuberculosis*, including cell wall synthesis, nucleic acid synthesis, and metabolic pathways, thereby inhibiting bacterial growth and survival.

Although these drugs significantly improved treatment outcomes, prolonged usage and improper administration led to selective pressure on the bacterium, ultimately contributing to the emergence of resistant strains.

1.3 Emergence of Drug-Resistant Tuberculosis

The emergence of drug resistance in *Mycobacterium tuberculosis* is primarily driven by genetic mutations in drug target genes, reduced drug permeability due to the complex cell wall, and the presence of efflux pump mechanisms. Multidrug-resistant TB (MDR-TB) and extensively drug-resistant TB (XDR-TB) pose serious challenges as they require longer, more toxic, and less effective treatment regimens. This significantly increases treatment failure rates and mortality.

MDR-TB is characterized by resistance to at least isoniazid and rifampicin, while XDR-TB shows additional resistance to second-line drugs, making treatment extremely difficult and less effective.

1.4 Need for Novel Drug Targets and Therapeutic Strategies

The rise of drug-resistant TB highlights the urgent need for new therapeutic agents and novel drug targets. Conventional anti-TB drugs require prolonged treatment durations (6–9 months or more), which increases the chances of treatment failure and resistance development. Additionally, these drugs may exhibit toxicity and reduced efficacy against resistant strains, along with poor patient compliance.

Therefore, identifying new molecular targets and developing drugs with improved efficacy, reduced toxicity, and shorter treatment duration has become a major focus of modern TB research.

Therefore, there is a critical need to identify novel molecular targets that are essential for bacterial survival but absent in humans to ensure selective toxicity. In this context, computational approaches such as structure-based drug design and virtual screening provide a powerful platform for rapid identification of potential inhibitors.

1.5 Targeting Cell Envelope Biosynthesis Pathways

Recent advances in molecular biology, structural biology, and bioinformatics have enabled the identification of several potential drug targets in *Mycobacterium tuberculosis*. Among these, enzymes involved in cell envelope biosynthesis are particularly promising.

Key components such as phosphatidylinositol mannosides (PIMs), lipomannan (LM), and lipoarabinomannan (LAM) are essential for maintaining cell structure, virulence, host–pathogen interactions, and antibiotic resistance. Disruption of these pathways can significantly affect bacterial survival.

1.6 Why cell wall is special in TB?

The mycobacterial cell envelope is highly complex and lipid-rich, contributing to intrinsic drug resistance and pathogenicity. Targeting enzymes involved in its biosynthesis can disrupt cell integrity, making it an attractive therapeutic strategy.

1.7 Role of Mannosyltransferases (PimA, PimB, PimC)

Phosphatidyl-myo-inositol mannosyltransferases such as PimA, PimB', and PimC play crucial roles in glycolipid biosynthesis.

PimA catalyses the transfer of a mannose moiety from GDP-mannose to phosphatidyl-myo-inositol (PI), forming phosphatidylinositol monomannoside (PIM), which is an essential initial step.

PimB' further adds a mannose residue to produce phosphatidylinositol dimannoside (PIM2).

PimC is involved in additional mannosylation steps, leading to higher-order PIMs and complex lipoglycans.

These glycolipids contribute to the structural integrity of the bacterial cell envelope and play an important role in immune modulation.

Among the various enzymes involved in cell envelope biosynthesis, phosphatidyl-myo-inositol mannosyltransferases such as PimA, PimB', and PimC play a crucial role in the formation of phosphatidylinositol mannosides (PIMs). These glycolipids serve as precursors for lipomannan (LM) and lipoarabinomannan (LAM), which are essential for cell wall integrity and host immune modulation. Due to their essentiality and absence in human metabolic pathways, these enzymes represent promising drug targets. In the present study, PimA and PimB' have been specifically selected for computational docking analysis to identify potential inhibitory compounds.

1.8 Importance of Pim Enzymes as Drug Targets

Since these enzymes are essential for bacterial survival and are absent in humans, they represent highly selective and promising targets for anti-TB drug discovery. Structural and computational studies, including molecular docking and molecular dynamics simulations, have provided insights into enzyme mechanisms and substrate interactions.

Such studies help identify key residues involved in catalysis and ligand binding, facilitating the rational design of novel inhibitors.

Although newer drugs such as bedaquiline, delamanid, and pretomanid have shown effectiveness against resistant TB strains, emerging resistance to these drugs has already been reported. This highlights the urgent need for continuous discovery of novel drug candidates.

1.9 Recent Advances in Anti-Tuberculosis Drug Development

Advancements in computational biology have revolutionized the drug discovery process. Techniques such as virtual screening, molecular docking, and molecular dynamics simulations enable rapid identification of potential drug candidates with high accuracy. Databases such as PubChem provide access to diverse chemical libraries, while docking tools like SwissDock facilitate prediction of protein–ligand interactions and binding affinities. In this study, these approaches have been employed to screen and evaluate potential inhibitors targeting Pim proteins.

10.0 Role of Computational Approaches in Drug Discovery

In summary, the increasing prevalence of drug-resistant tuberculosis necessitates the development of novel therapeutic strategies. Targeting essential enzymes involved in cell envelope biosynthesis, such as PimA and PimB', offers a promising approach. The integration of computational tools enables efficient screening and optimization of potential drug candidates. This study aims to contribute to the identification of novel inhibitors against *Mycobacterium tuberculosis*, thereby supporting the development of effective anti-TB therapies.

1.10 Computational Approach for Virtual Screening and Molecular Docking

In recent years, computational approaches have become integral to early-stage drug discovery, particularly for identifying potential inhibitors against mycobacterial targets. In the present study, a structure-based virtual screening strategy was employed to investigate ligand interactions with PimB' of *Mycobacterium smegmatis*. A set of 16 known ligands was selected, and their structural data were retrieved from PubChem using respective Compound IDs (CIDs). The ligand structures, initially obtained in SDF format, were converted into PDB format using Open Babel, followed by structural visualization and optimization in Avogadro, where they were further processed into MOL2 format for compatibility. To ensure appropriate force field parameters, ligands were subjected to parameterization using SwissParam under MMFF conditions in an aqueous environment. The optimized ligands were then docked against the target protein using SwissDock to predict binding modes and estimate binding affinities (ΔG values). The resulting docking conformations were evaluated based on energy scores and interaction stability, and the most favorable complexes were selected for further analysis. These protein–ligand complexes were subsequently prepared and converted into suitable formats for visualization, enabling detailed examination of binding orientation and interaction patterns. This integrated computational workflow provides a systematic framework for identifying promising ligand candidates for further experimental validation.

1.11 Conclusion and Research Significance

Targeting enzymes such as PimA and related mannosyltransferases involved in glycolipid biosynthesis represents a promising strategy for developing next-generation anti-tuberculosis drugs. Studying the structural dynamics and ligand interactions of these proteins can provide valuable insights into their catalytic mechanisms and aid in identifying potent inhibitors. Such research contributes significantly to combating drug-resistant tuberculosis and improving global health outcomes.

Methodology

2.1 Ligand Selection

A total of 16 known ligands were selected for the study based on their reported relevance. The structural and chemical information of these ligands was retrieved from PubChem using their respective Compound IDs (CIDs). These ligands were chosen to evaluate their binding potential with the target protein.

2.2 Ligand Structure Preparation

The three-dimensional (3D) structures of the selected ligands were downloaded in SDF format from PubChem. These structures were converted into PDB format using Open Babel to ensure compatibility with subsequent molecular modeling tools.

2.3 Ligand Optimization

The converted PDB files were imported into Avogadro for structural visualization and verification. The ligand structures were checked for any inconsistencies and subsequently converted into MOL2 format, which is suitable for parameterization and docking studies.

2.4 Parameterization of Ligands

The optimized ligand structures were uploaded to SwissParam to generate force field parameters. The parameterization was carried out using the Merck Molecular Force Field (MMFF) under aqueous (water) solvent conditions. The generated output files were downloaded, and the parameterized MOL2 files were extracted for further analysis.

2.5 Molecular Docking

Molecular docking studies were performed using SwissDock. The target protein (PimB' of *Mycobacterium smegmatis*) and the parameterized ligand structures were used as inputs. Docking simulations were conducted to predict possible binding modes and to estimate the binding affinity of ligands in terms of ΔG values.

2.6 Selection of Best Binding Conformations

The docking results generated multiple binding conformations for each ligand. These conformations were evaluated based on:

Binding free energy (ΔG)

Stability of ligand–protein interactions

The best-scoring conformation for each ligand was selected for further analysis.

2.7 Complex Preparation

The selected protein–ligand complexes corresponding to the best binding poses were prepared for visualization. These complexes were converted into suitable formats (such as PDB/PDBQT) to enable structural analysis using visualization tools.

2.8 Visualization and Analysis

The final docked complexes were visualized and analyzed using molecular visualization tools such as PyMOL. The binding orientation, positioning within the active site, and overall interaction patterns of the ligands with the protein were examined. Representative images of the docked complexes were generated for result interpretation.

Results

3.1 Molecular Docking Analysis

Molecular docking of the selected ligands with the target protein was performed using SwissDock. The docking process generated multiple binding conformations (clusters) for each ligand, representing different possible orientations within the binding site.

Each generated complex was associated with energy parameters, including FullFitness score and estimated binding free energy (ΔG), which were used to evaluate the stability and binding affinity of the ligand–protein interactions.

3.2 Selection of Top Docked Complexes

The obtained docking results were analyzed, and complexes were shortlisted based on:

Lowest FullFitness values

Most favorable (more negative) ΔG values

Complexes exhibiting better energy profiles were considered more stable and selected for further structural analysis. This step ensured that only the most promising ligand–protein interactions were taken forward.

3.3 Structural Preparation of Selected Complexes

The selected docked complexes were converted into PDB (Protein Data Bank) format to facilitate structural visualization and analysis. This conversion enabled compatibility with molecular visualization tools and allowed detailed inspection of ligand–protein interactions.

3.4 Visualization and Interaction Analysis

The prepared PDB structures were visualized using PyMOL. The spatial orientation of ligands within the binding pocket of the protein was examined.

Visualization revealed that the selected ligands occupied the active site region with favorable binding conformations. The interaction patterns, including positioning and fit within the binding pocket, were analyzed qualitatively.

3.5 Representation of Docked Complexes

Representative images of the selected ligand–protein complexes were generated using PyMOL and are presented below. These figures illustrate:

Binding orientation of ligands

Positioning within the active site

Overall ligand–protein interaction geometry

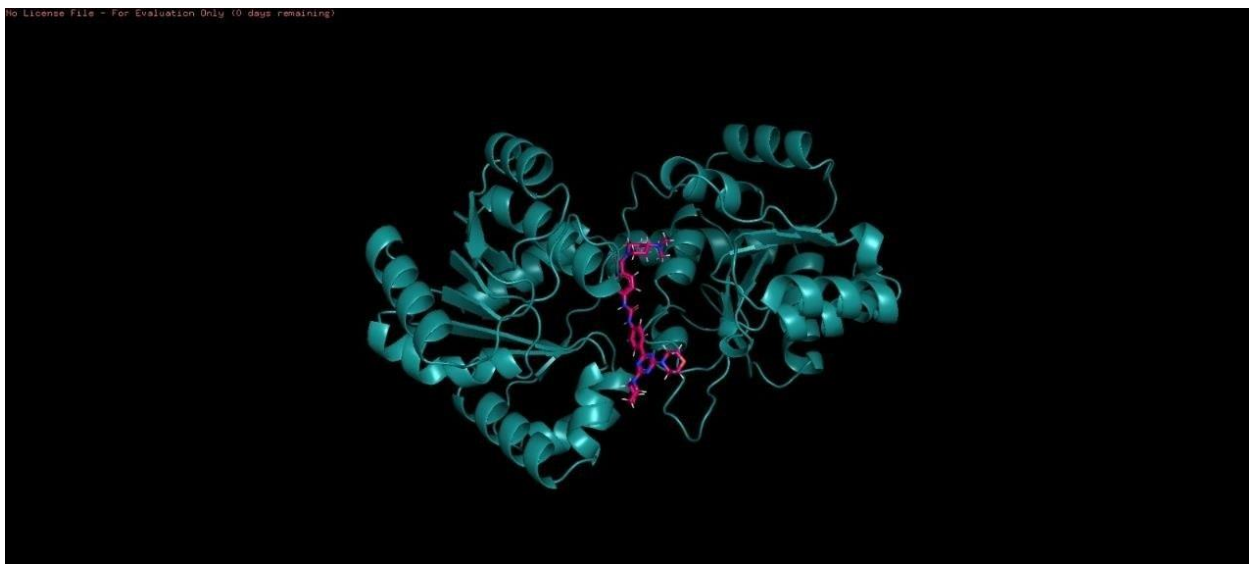


Figure 1: Docked complex of ligand 1 with PimB' showing binding orientation within the active site.

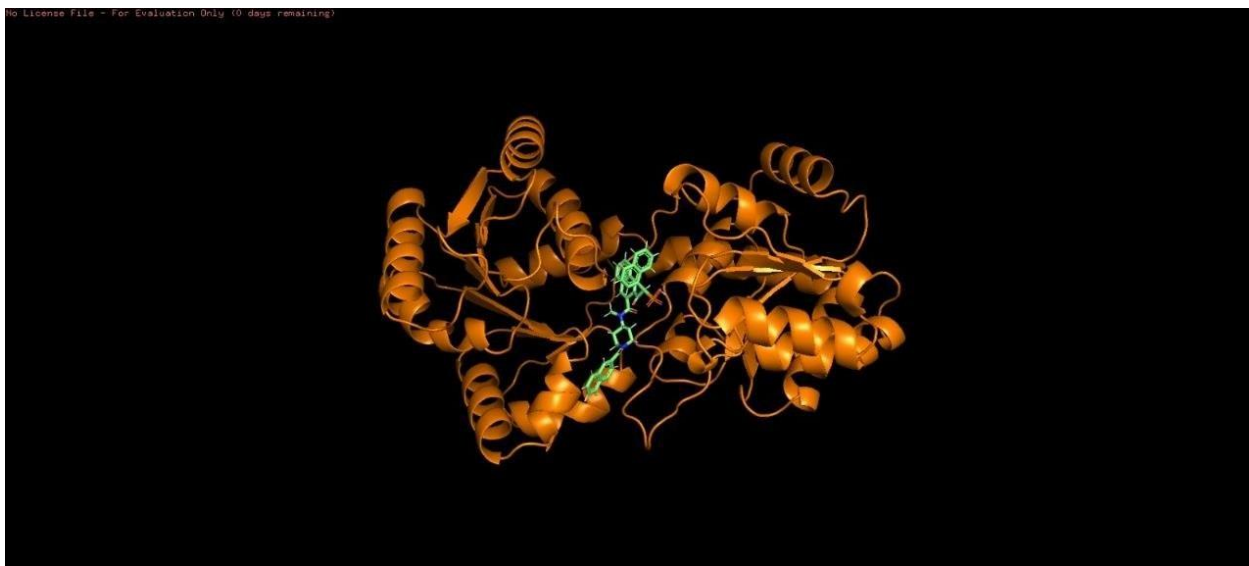


Figure 2: Docked complex of ligand 2 with PimB' showing binding orientation within the active site.

3.6 Summary of Docking Results

The docking study successfully identified several ligand–protein complexes with favorable binding energies and stable conformations. The selected complexes demonstrated promising

interaction profiles, suggesting their potential for further investigation in advanced computational and experimental studies.

Conclusion (Preliminary Statement)

This study presents a structured approach toward the identification of potential drug candidates against *Mycobacterium tuberculosis* using virtual screening and molecular docking techniques. However, as the literature review and detailed computational analyses are still ongoing, definitive conclusions cannot yet be drawn.

The outcomes of this project will be finalized after comprehensive evaluation of existing research, validation of selected targets, and thorough analysis of docking and ADMET results. Upon completion, the study is expected to provide meaningful insights into potential inhibitors and contribute to the development of effective anti-tubercular therapies.

This study successfully demonstrates the application of computational approaches in identifying potential drug candidates against *Mycobacterium tuberculosis*. Through virtual screening and molecular docking, a set of ligands was evaluated against key target proteins, PimA and PimB', which play a crucial role in cell envelope biosynthesis. The docking analysis revealed that several compounds exhibited strong binding affinity and stable interactions with the active sites of these enzymes.

The identified ligands showed favourable interaction profiles, including hydrogen bonding and hydrophobic interactions, which are essential for stable protein–ligand complex formation.

Overall, the study highlights the effectiveness of in silico drug design techniques in accelerating the drug discovery process. The results provide a strong foundation for further validation through molecular dynamics simulations and experimental studies. Targeting Pim enzymes represents a promising strategy for the development of novel anti-tubercular drugs, particularly in addressing the challenge of drug-resistant tuberculosis.

FUTURE SCOPE

The present study provides a computational basis for identifying potential inhibitors. However, further validation is required to establish their therapeutic applicability.

The top-ranked compounds identified through molecular docking will be shortlisted based on binding affinity and interaction profiles.

These selected complexes will be subjected to Molecular Dynamics (MD) simulations to evaluate the stability and conformational behavior of the ligand–protein interactions under dynamic conditions.

Binding free energies will be calculated using MM-GBSA and MM-PBSA approaches to obtain more accurate estimates of ligand affinity.

Subsequently, in vitro experimental validation will be performed to assess the biological activity of the compounds against *Mycobacterium tuberculosis*.

The target protein will be synthesized and purified using laboratory-based expression systems for biochemical assays.

Further studies will include evaluation of drug efficacy, toxicity, and pharmacokinetic behavior.

If necessary, lead optimization through structural modification will be carried out to enhance activity and reduce potential limitations.

Successful candidates may then proceed towards preclinical and clinical development stages.

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